

Emulator Superiority

When Machine Learning for PDEs Surpasses its Training Data

Felix Koehler & Nils Thuerey



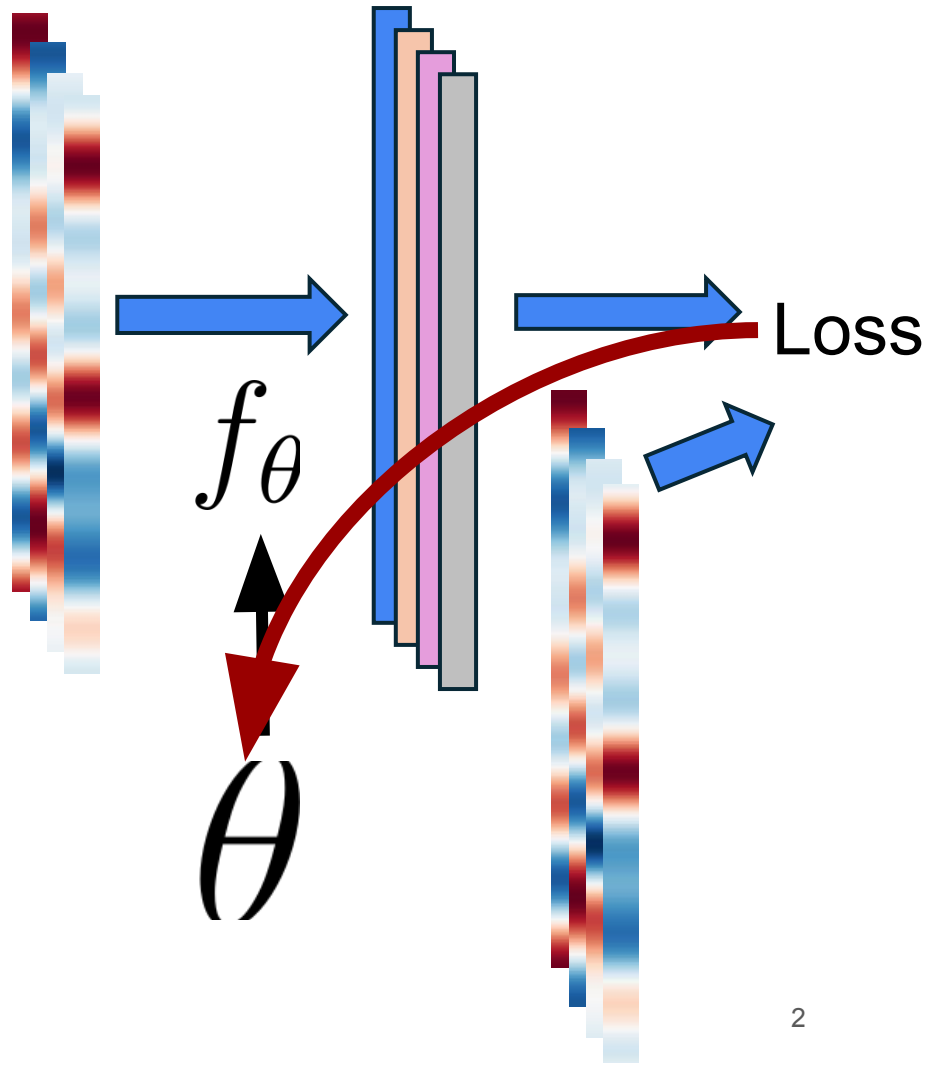
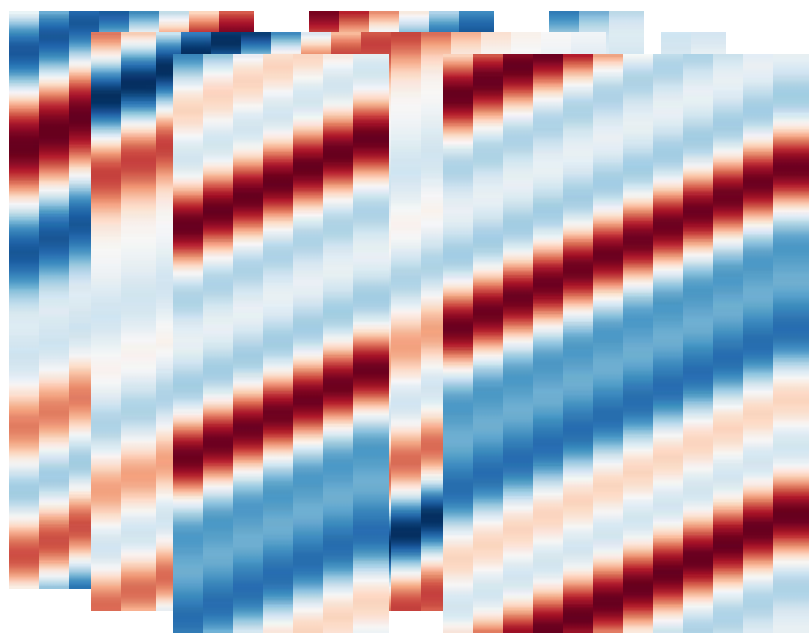
NEURAL INFORMATION
PROCESSING SYSTEMS



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Understanding Difference Between Training and Testing

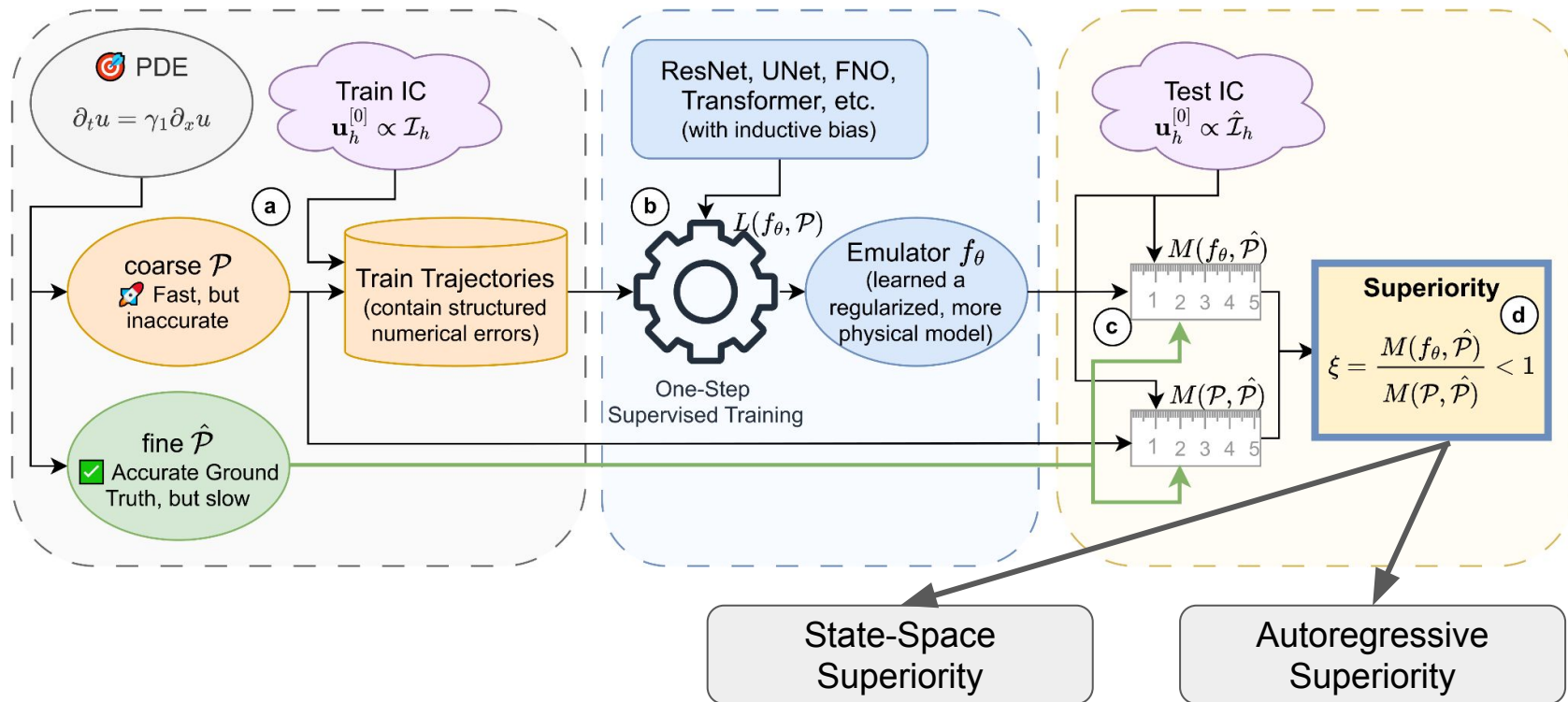
Training

$$L(\theta) = \mathbb{E}_{\{\mathbf{u}_h^{[t]}, \mathbf{u}_h^{[t+1]}\} \sim \mathcal{T}_{\mathcal{P}_h}(\mathbf{u}_h^{[0]}), \mathbf{u}_h^{[0]} \sim \mathcal{I}_h} \left[\zeta \left(f_{\theta}(\mathbf{u}_h^{[t]}), \mathbf{u}_h^{[t+1]} \right) \right]$$

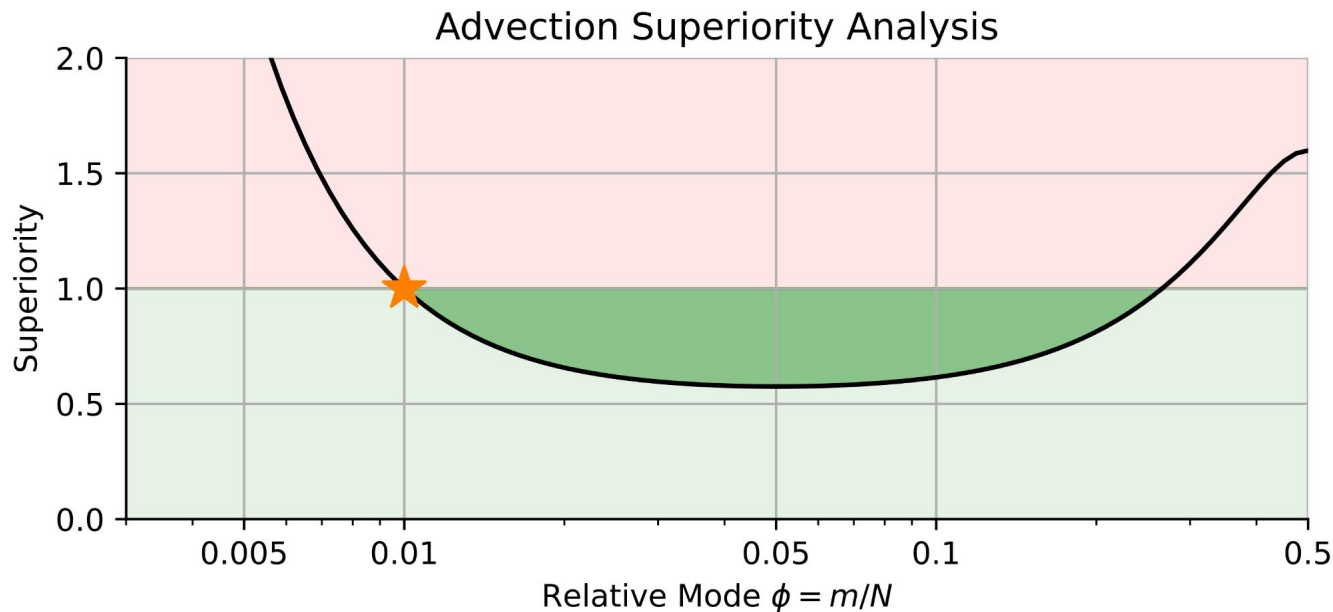
Testing

$$e^{[t]} = \mathbb{E}_{\mathbf{u}_h^{[0]} \sim \tilde{\mathcal{I}}_h} \left[\tilde{\zeta} \left(f_{\theta}^t(\mathbf{u}_h^{[0]}), \tilde{\mathcal{P}}_h^t(\mathbf{u}_h^{[0]}) \right) \right]$$

The missing piece: Where is the training data from?



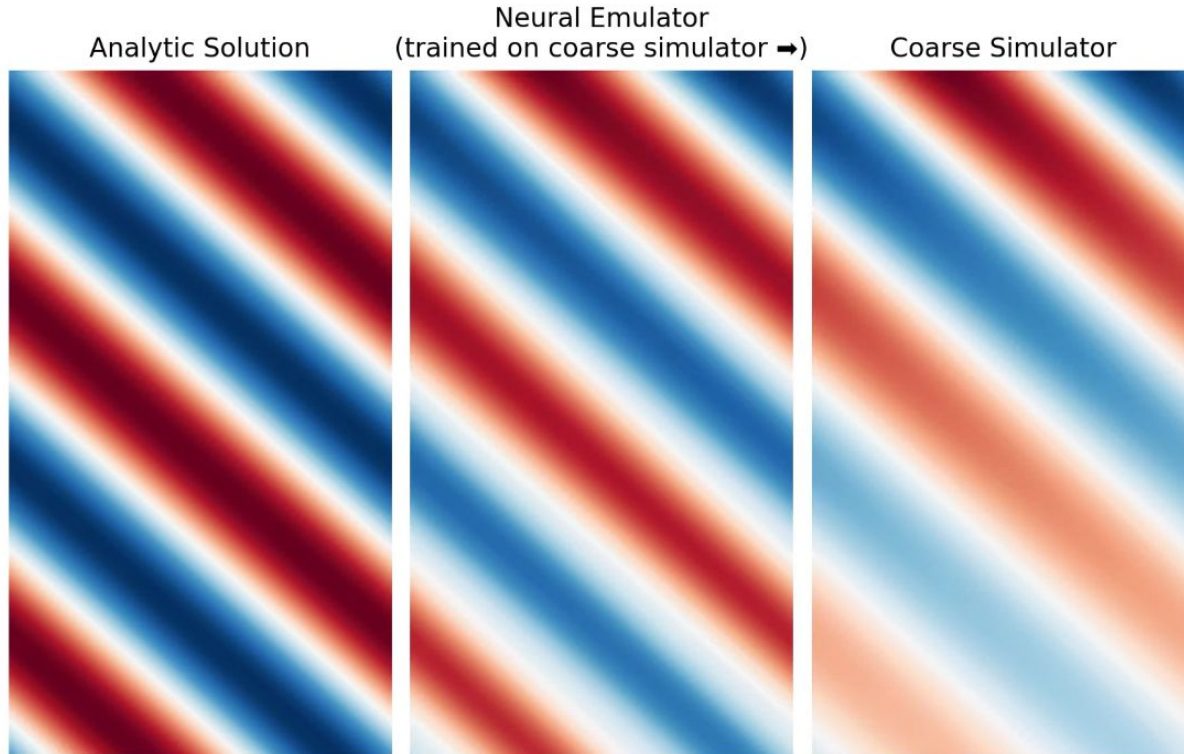
Theoretically Provable -> Forward State-Space Superiority



Nonlinear Emulators Achieve Autoregressive Superiority

Analytic Solution Neural Emulator
(trained on coarse simulator →) Coarse Simulator

Nonlinear Emulators Achieve Autoregressive Superiority



Neural Emulator Superiority: When Machine Learning for PDEs Surpasses its Training Data

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Fri 5 Dec 4:30 – 7:30 p.m.
Exhibit Hall C,D,E

tum-pbs.github.io/emulator-superiority